MIL-C-45854B

30 January 1973

SUPERSEDING
MIL-C-45854A(GL)
27 March 1968

MILITARY SPECIFICATION

CHASERS, THREAD, HAND (THREAD COMB)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hand thread chasers (see 6.1).

* 1.2 Classification. The hand thread chasers covered by this specification shall be of the following types and sizes, as specified (see 3.3.2, 6.2, and 6.3).

Type I - Internal

Type II - External

Size - 7 threads per inch.
Size - 12 threads per inch.
Size - 13 threads per inch.
Size - 14 threads per inch.
Size - 16 threads per inch.
Size - 18 threads per inch.
Size - 20 threads per inch.
Size - 24 threads per inch.
Size - 27 threads per inch.
Size - 28 threads per inch.
Size - 30 threads per inch.
Size - 32 threads per inch.
Size - 34 threads per inch.
Size - 36 threads per inch.
Size - 38 threads per inch.
Size - 40 threads per inch.
Size - 44 threads per inch.
Size - 48 threads per inch.

FSC 5136

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Size - 50 threads per inch.
Size - 56 threads per inch.
Size - 60 threads per inch.
Size - 64 threads per inch.
Size - 80 threads per inch.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-P-40 - Packaging and Packing of Hand Tools.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H-28 - Screw-Thread Standards for Federal Service

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

American Society for Testing and Materials (ASTM) Standard

E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies shall be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

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3. REQUIREMENTS

3.1 Standard product. The thread chasers covered by this specification shall be the current commercial product of the manufacturer, except for changes necessary to comply with the specification requirements.

3.2 Materials. The steel for the thread chasers shall be in accordance with the manfacturer's current commercial practice.

* 3.3 Design. The hand type I and type II thread chasers shall be designed for dressing damaged, stripped, or rusted right hand or left hand screw threads by a scraping or cutting action and shall be similar to figure 1. The tang shape and size shall be in accordance with the manufacturer's current commercial practice.

* 3.3.1 Teeth. The form of the chaser teeth shall be Unified Thread Form and shall comply with Handbook H28. The teeth shall have a Rockwell Hardness of C61-65 when tested as specified in 4.3.

* 3.3.2 Size. The thread chaser sizes shall be the tooth pitch, expressed as threads per inch.

3.4 Finish. All surfaces shall be finished in accordance with the manufacturer's commercial practice.

3.5 Marking. The tang of the thread chaser shall be permanently and legibly marked with the number of threads per inch that the chaser is to be used on and name or trademark of the manufacturer.

* 3.6 Workmanship. The hand chasers shall be free of slivers, nicks, rust, burrs, fractures or malformation. The chaser tang end shall be free of sharp edges. The chaser thread end shall not be bruised or broken.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

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* 4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

* 4.2.2 End item inspection. A lot shall consist of all thread chasers of one type and size offered for inspection at one time. The sample unit for this inspection shall be one completely fabricated thread chaser.

* 4.2.2.1 Visual examination. Examination shall be made for defects in table I. The inspection level shall be level II with acceptable quality level (AQL) of 2.5 for major defects and 4.0 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not in accordance with manufacturer's commercial practice.		X
Design	Not type or size specified.	X	
	Not provided with a tang.	X	
Workmanship	Slivers, nicks, or burrs.	X	
	Any rust areas.		X
	Fractured or malformed.	X	
	Sharp edges on tang.	X	
	Thread bruised or broken.	X	
Marking	Missing, illegible, not located as specified.		X

* 4.2.2.2 Thread form examination. Examination shall be made of the end item to determine compliance with 3.3.1 and 3.3.2. Any nonconformance shall be considered a defect. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of defects per hundred units.

* 4.2.2.3 Testing of the end item. The sample unit shall be tested in accordance with 4.3. Any nonconformance shall be considered a defect. The inspection level shall be S-3 with an AQL of 2.5, expressed in terms of defects per hundred units.

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* 4.2.3 Examination of preparation for delivery. An examination shall be made to determine that preservation, packaging, packing and marking comply with the requirements of PPP-P-40.

* 4.3 Hardness test. The thread form end of the hand chaser shall be tested for hardness in accordance with ASTM E18 to determine compliance with 3.3.1. Hardness shall be based on the average of three readings. The readings shall be taken on the flat face of the thread comb, 1/16 inch from the base line of the teeth, at a point on each side and center of the comb. Any nonconformance shall constitute failure to pass this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking for level A, B, or C shall be in accordance with PPP-P-40.

* 6. NOTES

6.1 Intended use. The hand thread chasers covered by this specification are intended to be used for dressing damaged screw threads by a scraping or cutting action.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and size required (see 1.2).
(c) Selection of the applicable levels of preservation and packaging, and packing (see 5.1).

* 6.3 Classification. Sequential numbering of sizes (1 thru 10), appearing in previous issues of this specification, has not been used in this revision since it is considered superfluous.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:

Army - GL
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Air Force - 84

Review activities:

Army - AV, WC, ME

User activities:

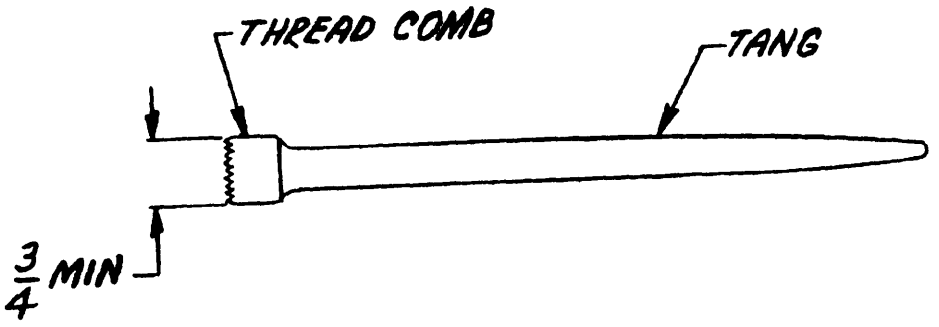
Army - CE, EL, AT
Navy - MC

Preparing activity:

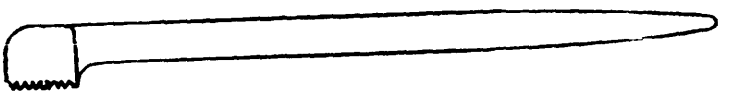
Army - GL

Project No. 5136-0032

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TYPE I-EXTERNAL THREADS



TYPE II-INTERNAL THREADS

FIGURE 1. CHASER, THREAD, HAND(THREAD COMB)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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